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FRIDAY, APRIL 20, 2012

Digital design Interview Questions

1. *If inverted output of D flip-flop is connected to its input how the flip-flop behaves?*
2. *Design a circuit to divide input frequency by 2?*
3. *Design a divide by two counter using D-Latch.*
4. *Design a divide-by-3 sequential circuit with 50% duty cycle.*
5. *What are the different types of adder implementation?*
6. *Draw a Transmission Gate-based D-Latch?*
7. *Give the truth table for a Half Adder. Give a gate level implementation of the same.*
8. *Design an OR gate from 2:1 MUX.*
9. *What is the difference between a LATCH and a FLIP-FLOP?*
10. *Design a D Flip-Flop from two latches.*
11. *Design a 2 bit counter using D Flip-Flop.*
12. *What are the two types of delays in any digital system*
13. *Design a Transparent Latch using a 2:1 Mux.*
14. *Design a 4:1 Mux using 2:1 Mux's.*
15. *What is metastable state? How does it occur?*
16. *What is metastablity?*
17. *Design a 3:8 decoder*
18. *Design a FSM to detect sequence "101" in input sequence*
19. *Convert NAND gate into Inverter in two different ways.*
20. *Design a D and T flip flop using 2:1 mux only.*
21. *Design D Latch from SR flip-flop.*
22. *Define Clock Skew, Negative Clock Skew, Positive Clock Skew?*
23. *What is race condition? How it occurs? How to avoid it?*
24. *Design a 4 bit Gray Counter?*
25. *Design 4-bit synchronous counter, asynchronous counter?*
26. *Design a 16 byte asynchronous FIFO?*
27. *What is the difference between a EEPROM and FLASH?*
28. *What is the difference between a NAND-based Flash and NOR-based Flash?*
29. *Which one is good: asynchronous reset or synchronous reset? Why?*
30. *Design a simple circuit based on combinational logic to double the output frequency.*
31. *What is the difference between flip-flop and latch?*
32. *Implement comparator using combinational logic, that compares two 2-bit numbers A and B. The comparator should have 3 outputs: A > B, A < a =" B.">*
33. *Give two ways of converting a two input NAND gate to an inverter?*
34. *What is the difference between mealy and moore state-machines?*
35. *What is the difference between latch based design and flip-flop based design?*
36. *What is metastability and how to prevent it?*
37. *Design a four-input NAND gate using only two-input NAND gates.*
38. *Why are most interrupts active low?*
39. *How do you detect if two 8-bit signals are same?*
40. *7 bit ring counter's initial state is 0100010. After how many clock cycles will it return to the initial state?*
41. *Design all the basic gates NOT, AND, OR, NAND, NOR, XOR, XNOR using 2:1 Multiplexer.*
42. *How will you implement a full subtractor from a full adder?*
43. *In a 3-bit Johnson's counter what are the unused states?*
44. *What is difference between RAM and FIFO?*
45. *What is an LFSR? List a few of its industry applications.*
46. *Implement the following circuits:*
47. *(a) 3 input NAND gate using minimum number of 2 input NAND gates*
48. *(b) 3 input NOR gate using minimum number of 2 input NOR gates*
49. *(c) 3 input XNOR gate using minimum number of 2 input XNOR gates assuming 3 inputs A,B,C?*
50. *Design a D-latch using (a) using 2:1 Mux (b) from S-R Latch?*
51. *How to implement a Master Slave flip flop using a 2 to 1 mux?*
52. *How many 2 input xor's are needed to inplement 16 input parity generator?*
53. *Convert xor gate to buffer and inverter.*

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54. *Difference between onehot and binary encoding?*
55. *What are different ways to synchronize between two clock domains?*
56. *How to calculate maximum operating frequency?*
57. *How to find out longest path?*
58. *How to achieve 180 degree exact phase shift?*
59. *What is significance of ras and cas in SDRAM?*
60. *Tell some of applications of buffer?*
61. *Implement an AND gate using mux?*
62. *What will happen if contents of register are shifter left, right?*
63. *What is the basic difference between analog and digital design?*
64. *What advantages do synchronous counters have over asynchronous counters?*
65. *What types of flip-flops can be used to implement the memory elements of a counter?*
66. *What are the advantages of using a microprocessor to implement a counter rather than the conventional method (flip-flop and logic gates)?*
67. *What is the principal advantage of Gray Code over straight (conventional) binary?*
68. *What does Pipelining do?*
69. *Design divide by 2, divide by 3 circuit with equal duty cycle.*
70. *How many 4:1 mux do you need to design a 8:1 mux?*
71. *What is D-Word, Q-word?*
72. *Define Moore, Mealy state machines. Which one is good for timing?*
73. *Design a FSM to detect 10110. What is the minimum number of flops required?*
74. *Design a simple circuit based on combinational logic to double the output frequency.*
75. *Design a 2bit up/down counter with clear using gates. (No verilog or vhdl)*
76. *Design a finite state machine to give a modulo 3 counter when x=0 and modulo 4 counter when x=1.*
77. *Minimize: S= A' + AB*
78. *What is the function of a D-flipflop, whose inverted outputs are connected to its input?*
79. *How to synchronize control signals and data between two different clock domains?*
80. *Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.*
81. *In what cases do you need to double clock a signal before presenting it to a synchronous state machine?*
82. *How many bit combinations are there in a byte?*
83. *What are the different Adder circuits you studied?*
84. *Give the truth table for a Half Adder. Give a gate level implementation of the same.*
85. *Convert 65(Hex) to Binary*
86. *Convert a number to its two's compliment and back.*
87. *What is the 1's and 2's complement of the decimal number 25.*
88. *If A?B=C and C?A=B then what is the boolean operator ?*

Posted by Kumar at 7:45 AM  

1 comment:



Anonymous June 3, 2012 12:37 PM

is there any chance i can find the answers to these questions?
i'm a student in desperate condition...i have exam tomorrow and i don't know how to solve the questions involving MUX , Latches and flip-flops

any help??
can anybody at least give me hints?

my best regards in advance.

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